

 **game (1)(2).py**
Code **game(5).py**
Code **README(1).md**
Document **Lab_4_VibeCoding...**
PDF

i5t is my se lab

start from first

Microsoft Windows [Version 10.0.26100.4946]
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```
C:\Users\Rushikesh Patil>
C:\Users\Rushikesh Patil>cd
"$env:USERPROFILE\Downloads\lunar_lander"
The filename, directory name, or volume label syntax is incorrect.
```

```
C:\Users\Rushikesh Patil>dir
Volume in drive C is Windows
Volume Serial Number is 463B-D231
```

Directory of C:\Users\Rushikesh Patil

07-10-2026	21:49	<DIR>	.
04-10-2026	16:26	<DIR>	..
05-04-2026	16:22	<DIR>	.arduinoIDE
25-09-2026	22:28	<DIR>	.cache
09-10-2026	20:10	<DIR>	.codex
04-10-2026	15:18	<DIR>	.config
04-10-2026	17:09	<DIR>	.copilot
04-10-2026	19:49	<DIR>	.docker
26-07-2026	19:01	70	.gitconfig
26-07-2026	11:56	<DIR>	.ipython
26-07-2026	12:16	<DIR>	.jupyter
26-09-2026	00:10	<DIR>	.matplotlib
26-09-2026	01:12	<DIR>	.sbx-denybin
21-04-2026	13:12	<DIR>	.VirtualBox
05-10-2026	13:18	<DIR>	.vscode

```

09-06-2026 19:08 <DIR>      .vscode-shared
24-06-2025 18:12 <DIR>      ansel
14-08-2026 08:39      141,540 battery-report.html
22-05-2025 23:11 <DIR>      Contacts
04-09-2026 10:49 <DIR>      CrossDevice
04-10-2026 15:14 <DIR>      Desktop
02-10-2026 14:54 <DIR>      Documents
09-10-2026 20:10 <DIR>      Downloads
19-08-2026 21:35 <DIR>      driver_demo
22-05-2025 23:11 <DIR>      Favorites
19-08-2026 21:28 <DIR>      feast-env
06-11-2025 08:48      0 git
11-12-2024 16:02      3,782 import csv.py
22-05-2025 23:11 <DIR>      Links
22-05-2025 23:11 <DIR>      Music
16-09-2025 12:08 <DIR>      my-app
09-10-2026 19:06 <DIR>      OneDrive
09-03-2025 11:02 <DIR>      PES1UG24AM394
10-03-2026 22:16 <DIR>      Pictures
16-09-2026 15:00 <DIR>      predict_srn
24-11-2024 18:52      8 python.txt
22-05-2025 23:11 <DIR>      Saved Games
22-05-2025 23:11 <DIR>      Searches
27-09-2026 00:31 <DIR>      strikecast-frontend
18-01-2025 18:59      45 Untitled-1.py
09-10-2026 18:13 <DIR>      Videos
04-02-2026 09:40 <DIR>      VirtualBox VMs
04-10-2026 19:46 <DIR>      workplace-compliance-desk
07-10-2026 21:49 <DIR>      WPS Cloud
        6 File(s)      145,445 bytes
        38 Dir(s) 61,240,631,296 bytes free

```

C:\Users\Rushikesh Patil>

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C:\Users\Rushikesh Patil\Downloads\lunar_lander>

C:\Users\Rushikesh Patil\Downloads\lunar_lander>cd /d

"%USERPROFILE%\Downloads\lunar_lander"

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>
C:\Users\Rushikesh Patil\Downloads\lunar_lander>dir
Volume in drive C is Windows
Volume Serial Number is 463B-D231
```

Directory of C:\Users\Rushikesh Patil\Downloads\lunar_lander

```
09-10-2026 20:05 <DIR>      .
09-10-2026 20:10 <DIR>      ..
09-10-2026 20:05      7,240 game (1).py
09-10-2026 19:56      7,191 game.py
09-10-2026 19:53      4,028 README.md
                3 File(s)      18,459 bytes
                2 Dir(s) 61,239,263,232 bytes free
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>
```

yes game will opened



20261009-1447-33....
File

keep this video save for you

 Memory updated

```
lunar_lander>
C:\Users\Rushikesh Patil\Downloads\lunar_lander>python -m pip
show pygame
Name: pygame
Version: 2.6.1
Summary: Python Game Development
Home-page: https://www.pygame.org
Author: A community project.
Author-email: pygame@pygame.org
License: LGPL
Location: C:\Users\Rushikesh
Patil\AppData\Local\Programs\Python\Python311\Lib\site-
packages
Requires:
```

Required-by:

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>
C:\Users\Rushikesh Patil\Downloads\lunar_lander>
C:\Users\Rushikesh Patil\Downloads\lunar_lander>python
game.py
pygame 2.6.1 (SDL 2.28.4, Python 3.11.9)
Hello from the pygame community.
https://www.pygame.org/contribute.html
```



```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>
C:\Users\Rushikesh Patil\Downloads\lunar_lander>python
game.py
File "C:\Users\Rushikesh Patil\Downloads\lunar_lander\game.py",
line 90
    self.lives -= 1
                ^
IndentationError: unindent does not match any outer indentation
level
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>
```



20261009-1453-53....
File

after fixing task1

yes find it in notepad

yes game open sussefully

yes

yes



20261009-1502-14....

File

yes after complete all task changing colour when fuel dearese

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>python -m  
py_compile game.py
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>git status  
fatal: not a git repository (or any of the parent directories): .git
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>
```

On branch master

No commits yet

Untracked files:

(use "git add <file> ..." to include in what will be committed)

```
README.md  
__pycache__/  
game (1).py  
game.py
```

nothing added to commit but untracked files present (use "git
add" to track)

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>
```

nothing added to commit but untracked files present (use "git
add" to track)

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>echo  
__pycache__ / > .gitignore
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>echo *.pyc >>  
.gitignore
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>echo .venv/ > >  
.gitignore
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>git status  
On branch master
```

```
No commits yet
```

```
Untracked files:
```

```
(use "git add <file> ..." to include in what will be committed)
```

```
.gitignore  
README.md  
game (1).py  
game.py
```

```
nothing added to commit but untracked files present (use "git  
add" to track)
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>
```

```
I already created my GitHub repository. Its URL is:
```

```
https://github.com/Rushikesh9742/PES1UG24AM394\_SE\_LAB
```

```
nothing added to commit but untracked files present (use "git  
add" to track)
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>git remote add  
origin  
https://github.com/Rushikesh9742/PES1UG24AM394\_SE\_LAB.git
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>git remote -v  
origin  
https://github.com/Rushikesh9742/PES1UG24AM394\_SE\_LAB.git  
(fetch)  
origin  
https://github.com/Rushikesh9742/PES1UG24AM394\_SE\_LAB.git  
(push)
```

C:\Users\Rushikesh Patil\Downloads\lunar_lander>



```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>git fetch origin
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (10/10), done.
remote: Total 12 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (12/12), 1.27 MiB | 2.41 MiB/s, done.
From
https://github.com/Rushikesh9742/PES1UG24AM394_SE_LAB
* [new branch]    main    -> origin/main
```

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>git branch -a
remotes/origin/HEAD -> origin/main
remotes/origin/main
```

C:\Users\Rushikesh Patil\Downloads\lunar_lander>

```
C:\Users\Rushikesh Patil\Downloads\lunar_lander>type
README.md
# Lunar Lander Repair Lab
```

This project is a single-file Lunar Lander-lite clone using **Pygame**. It introduces students to vector physics, procedural terrain generation, and multi-condition landing validation using a small, readable object-oriented codebase.

What's Provided

A working Lunar Lander-lite game with:

- A ship with rotation, thrust, gravity, and limited fuel, wrapping

horizontally around the screen

- Procedurally generated terrain with two landing pads of different point multipliers
- Touchdown validation based on speed and angle, with a full HUD showing live telemetry
- Levels, lives, and scoring for successful landings

It has **one deliberate bug** and **three optional features** left as empty functions. You are expected to **analyze**, **interact** with an AI assistant, and **complete/fix** the game to make it fully functional and more interesting.

Use an LLM (e.g. ChatGPT or Claude) as your debugging and pair-programming partner for this lab.

Getting Started

Setup

1. Make sure you have Python 3.10+ installed.
2. Install dependencies:

```
bash  
  
pip install pygame
```

3. Run the game:

```
bash  
  
python game.py
```

Controls: Left/Right to rotate, Up to thrust, R to reset, Space to continue after landing or crashing.

Tasks to Complete

Each task must be completed using an iterative process involving LLM suggestions and your critical code review.

Task 1: Fix the touchdown validation bug

> A safe landing is supposed to require the ship to be over a pad, descending slowly, roughly level, *and* not moving sideways too fast. In the current build, a ship can slam down sideways ΓÇö well beyond MAX_SPEED_X ΓÇö on a pad at a safe vertical speed and angle, and it still counts as a perfect landing. Check the condition in `touchdown()` against all three constants it defines near the top of the file (`MAX_SPEED_X`, `MAX_SPEED_Y`, `MAX_ANGLE`), and see which one it never actually checks.

Task 2: Implement `ship_color(fuel_ratio)`

> Called once per frame in `draw`, as `color = ship_color(max(0.0, self.fuel) / FUEL_MAX)` or `(230, 230, 240)`. It receives the remaining fuel as a fraction from 0.0 (empty) to 1.0 (full) and should return an `(r, g, b)` hull color, or `None` to keep the default. Idea: shift the hull toward red as fuel runs low.

Task 3: Implement `on_landing(score)`

> Called from `touchdown()` immediately after a successful landing's points are added to the score. It receives the number of points just earned from that landing. Its return value is ignored. Idea: a fireworks effect, or a distinct sound for a high-multiplier pad.

Task 4: Implement `bonus_life_threshold()`

> Called every frame in `update()`. It takes no arguments and should return an integer score value, or `None` to disable bonus lives entirely. Whenever the score crosses a multiple of that value

for the first time, one life is awarded automatically ΓÇö the bookkeeping (self.bonus_awarded) is already implemented, so you only need to choose the threshold. Idea: return 1500.

Expected Behavior

- The ship wraps horizontally but never moves past the top of the screen; gravity and rotated thrust behave like real vectors
- Terrain always includes exactly two landing pads with different score multipliers
- A landing only counts as successful when the ship is over a pad, descending slowly, roughly level, and not drifting sideways too fast
- Fuel never goes negative, and thrust cuts out automatically once it runs out
- A crash costs a life; the game ends when lives reach zero

Folder Structure

```
lunar_lander/
Γö£ΓöÇΓöÇ game.py
ΓööΓöÇΓöÇ README.md
```

Submission Checklist

Submission is only the following three things:

- [] A 10-second video of gameplay ****before**** your changes, showing the bug/broken behavior
- [] A 10-second video of gameplay ****after**** your changes, showing the bug fixed and the new features working

- [] The Chat/LLM used page link, with the complete chat history

C:\Users\Rushikesh Patil\Downloads\lunar_lander>

Submission is only the following three things:

- [] A 10-second video of gameplay ****before**** your changes, showing the bug/broken behavior
- [] A 10-second video of gameplay ****after**** your changes, showing the bug fixed and the new features working
- [] The Chat/LLM used page link, with the complete chat history

C:\Users\Rushikesh Patil\Downloads\lunar_lander>git switch -c main --track origin/main

error: The following untracked working tree files would be overwritten by checkout:

README.md

Please move or remove them before you switch branches.

Aborting

C:\Users\Rushikesh Patil\Downloads\lunar_lander>git switch -c main --track origin/main

error: The following untracked working tree files would be overwritten by checkout:

README.md

Please move or remove them before you switch branches.

Aborting

C:\Users\Rushikesh Patil\Downloads\lunar_lander>

🌐 <https://chatgpt.com/c/6ac8fd41-a444-83ee-a390-353225ff12fc>

next step

task2

i done all 4

all task in 2videos coverd

next step

My SE Lab 4 submission location is: GitHub repository. Guide me step by step to submit the required two videos and ChatGPT conversation link.



Lab_4_VibeCoding...
PDF

you only give file in that cover both video final file before that you check this according to that give

Worked for 20s

Analyzed

</>

yes i downloaded zip file